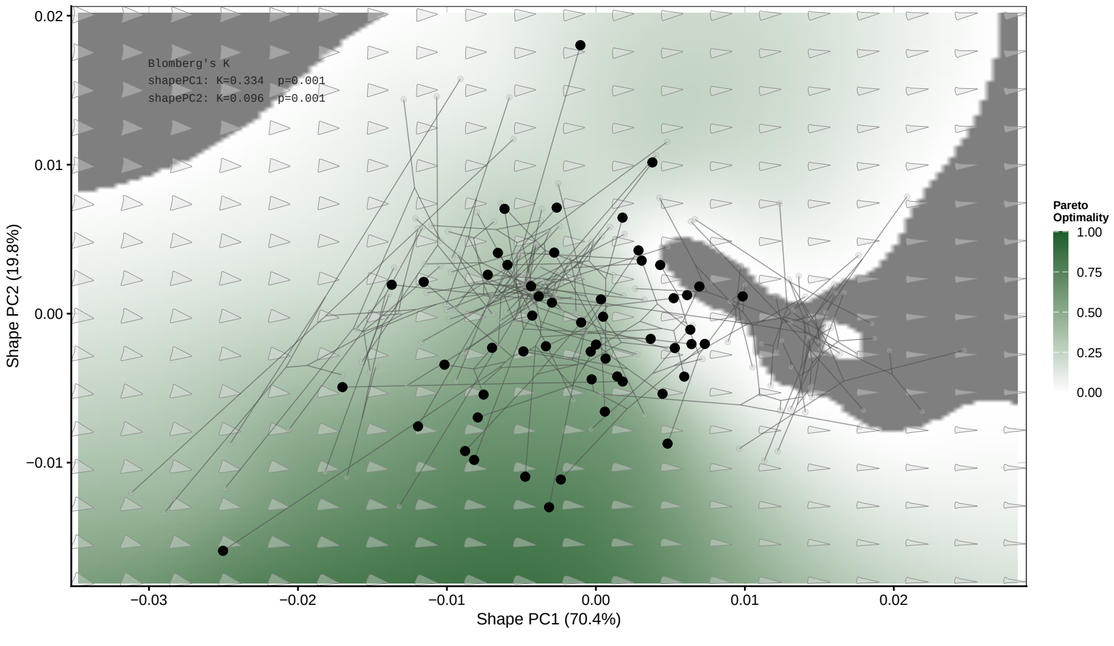
a

b

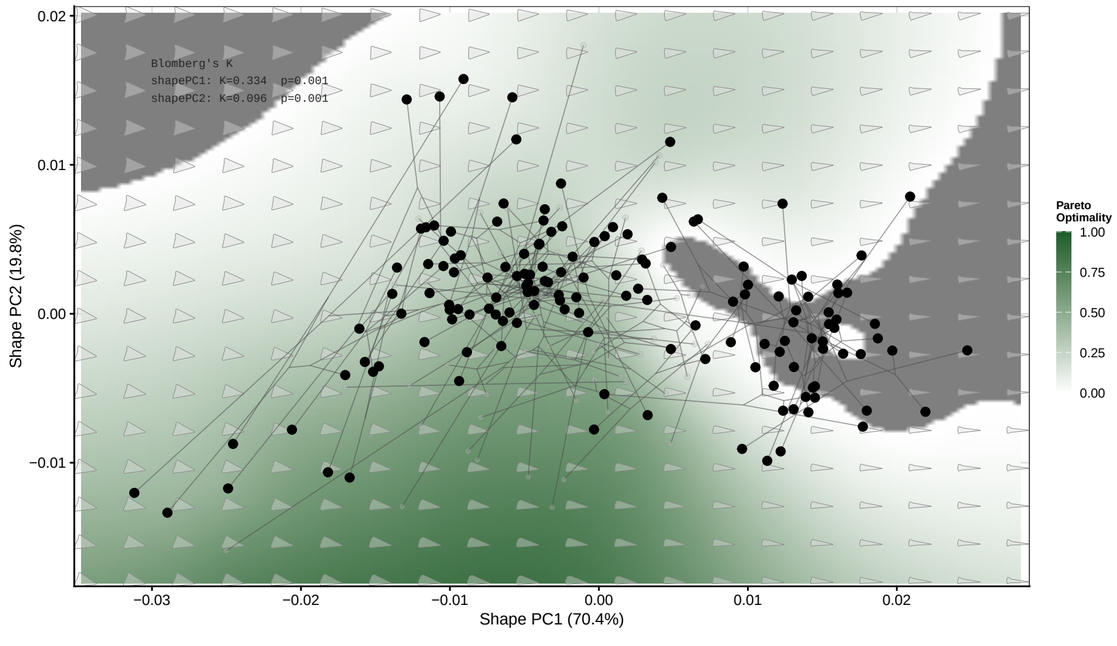
(a) Non−Pterodactyliformes Phylomorphospace

Anurognathidae, Rhamphorhynchidae, Non−breviquartossan,
Basal pterosaur − Pareto optimality landscape



(b) Pterodactyloidea Phylomorphospace

Derived pterosaurs projected onto optimality landscape



c

(c) Pareto Optimality by Grade

Non−Pterodactyliformes vs Pterodactyliformes

